

## Thin chip-free biodegradable tags monitor temperature and humidity

Researchers at Empa, EPFL, and CSEM in Switzerland developed a fully biodegradable, chip-free smart tag that



The biodegradable tag is as thin as a sheet of paper, but still able to measure the temperature and relative humidity (Image: Empa)

wirelessly monitors temperature and humidity without batteries or silicon. Printed on compostable biopolymer and cellulose, the tag's resonance shifts with environmental conditions and permanently flags excursions above thresholds, enabling sustain-

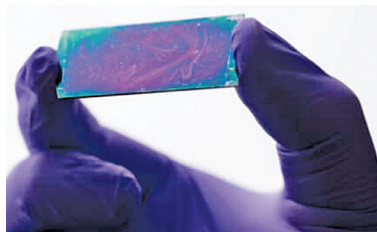
able cold-chain monitoring for food, vaccines, and perishable goods, reducing e-waste from disposable sensors.

## India hits 1Tbps milestone across 1450km fibre route

New Delhi-Mumbai/Chennai telecom corridor boosts India's digital backbone as network operators hit a 1Tbps transmission milestone over a long-haul fibre route that supports hyperscalers, cloud services, and AI data flows. Constl, using Ciena's WaveLogic 6 Extreme coherent optics, transported a full terabit per second across 1450km without regeneration, cutting latency and operational cost while doubling wavelength capacity and improving spectral efficiency for next-gen connectivity.

## A thin film that helps AI processors run cooler, with added benefits

A thin-film thermal regulator for chips helps AI processors run cooler, improving performance and energy efficiency for



This is the two-dimensional thin film electric insulator designed in a University of Houston lab to make AI faster and reduce power consumption (Credit: <https://www.egr.uh.edu>)

data centres, edge AI devices, and high-performance computing. The film developed by researchers at the University of Houston passively dissipates heat across chip surfaces, reducing hot-spots and throttling, enabling longer, sus-

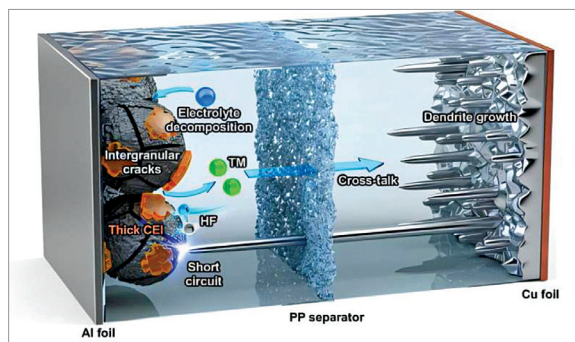
tained workloads without additional cooling infrastructure. This approach can lower operational costs and enhance reliability for large-scale AI training, inference engines, and embedded intelligence applications.

## Wide-angle, glasses-free 3D visuals brought to desktop-size monitors

An AI-driven light-field display system called EyeReal, developed by researchers in Shanghai, China, brings wide-angle, glasses-free 3D visuals to desktop-scale monitors, eliminating the need for special eyewear by tracking eye/head position and computing only the light rays needed for the viewer's perspective. This boosts immersive 3D for gaming, design visualisation, education and industrial graphics while avoiding bulky optics, widening viewing angles beyond 100° with natural motion parallax.

## Batteries that last longer and charge safer

A new lithium-ion battery design by researchers at POSTECH, Gyeongsang National University, and Korea Institute of Energy Research (KIER) improves longevity and



Schematic illustrations of a conventional polypropylene (PP) separator (top) and the molecularly engineered separator (MFS, bottom) in lithium-metal batteries (Credit: POSTECH)

safety for EVs, consumer electronics, and grid storage by minimising lithium dendrite growth and enhancing thermal stability. Using a reinforced anode and safer electrolyte, the batteries maintain capacity over more cycles while reducing the risk of short circuits or thermal runaway. This advance could extend device lifespans, cut maintenance costs, and make high-performance energy storage more reliable across applications.

## Laser-driven 3D printing for free-standing thermoset structures

Researchers at Xiamen University (Xiamen, China) have developed a new laser-driven 3D printing method that rapidly builds free-standing thermoset structures without support material, enabling precise, programmable architectures for flexible electronics, soft robotics, microfluidic systems, and biomedical scaffolds. By curing polymer jets with a focused laser in under 0.25s and tuning mechanical/electrical properties on the fly, manufacturers could cut turnaround time and expand multifunctional device designs in wearable tech and complex components.